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Discriminating B and T-lymphocyte from its molecular profile acquired in a label-free and high-throughput method

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ABSTRACT

B and T-lymphocytes are one of the main players of the adaptive immune system and consequently, the ability to discriminate these cells is relevant in diverse pathophysiological cases. In the present work, FTIR spectroscopy was used to acquire the cells molecular profile in a label-free, simple, rapid, economic, and high-throughput mode. It was possible, by hierarchical cluster analysis of the second derivative spectra, between 900 to 1500 cm^{-1}, to correctly classify 98 % of samples as B and T-lymphocyte. It was also possible to develop a very good support vector machine model, which could predict B and T-lymphocyte from the second derivative spectra, with high accuracy (0.95), precision (0.95), recall (0.95), and F1 score (0.95). The method developed presents therefore, appealing characteristics to promote a new diagnostic tool to analyze and discriminate B from T-lymphocytes.

1. Introduction

The immune system is composed of a multiplicity of organs and cells. These act in an integrated mode towards the defense of the host against molecular patterns, present in foreign substances, i.e. antigens, such as bacteria and altered host cells due e.g. to viral infections and carcinogenesis. The adaptive immune system, also known as acquired immunity, complements the innate immune system, by enabling the creation of a high diversity of lymphocyte cells. These lymphocytes facilitate the recognition of a much higher diversity of antigen signatures, and consequently, more specific than those recognized by the innate system. Therefore, the specific system enables the immune system to adapt the response towards pathogens that present a high diversity of patterns [1].

Lymphocytes play a key role in acquired immunity. They can be divided into several major cell subsets with multiple phenotypes, such as Natural Killer, B, and T lymphocytes. These cells act in a highly discriminatory manner and thus displaying specificity. Most B lymphocytes specific functions are mediated by its secreted proteins, antibodies [1], whereas T-lymphocytes present a higher diversity of cell types and consequently cellular functions: cytotoxic T-lymphocytes, that act against the host cells that were altered by infection processes or due to carcinogenic process, and a high diversity of T-helper cells that are major regulators of the whole immune system [1].

The dysregulation of specific immune responses, as B and T-lymphocyte responses, is associated with diverse diseases as infections, auto-immune, allergies, cancer progression, and even in rejection of transplanted organs. Therefore, there are diverse situations by which analysis of B and T-lymphocytes are clinically relevant. For example, in immune rejection of transplanted organs, it is paramount to define the specific immune rejection mechanism that can be mediated through the production of antibodies or cytotoxic T-lymphocytes [2–4]. In carcinogenesis, it is important to define the tumor infiltrating-lymphocytes, since these cells may affect either the disease prognosis or the immunotherapy treatment itself [5,6]. In both these examples, the characterization of the B and T immune cells will promote a more precise diagnosis and prognosis, enabling to better define, monitor and control the expensive immunotherapies. Ultimately, the monitoring of these immunological processes will also promote the knowledge of the intricate and complex immune network towards either the disease development or the immunotherapy itself, contributing to better diagnosis, prognosis and therapies [7,8].

Current methodologies to identify lymphocyte subpopulations rely

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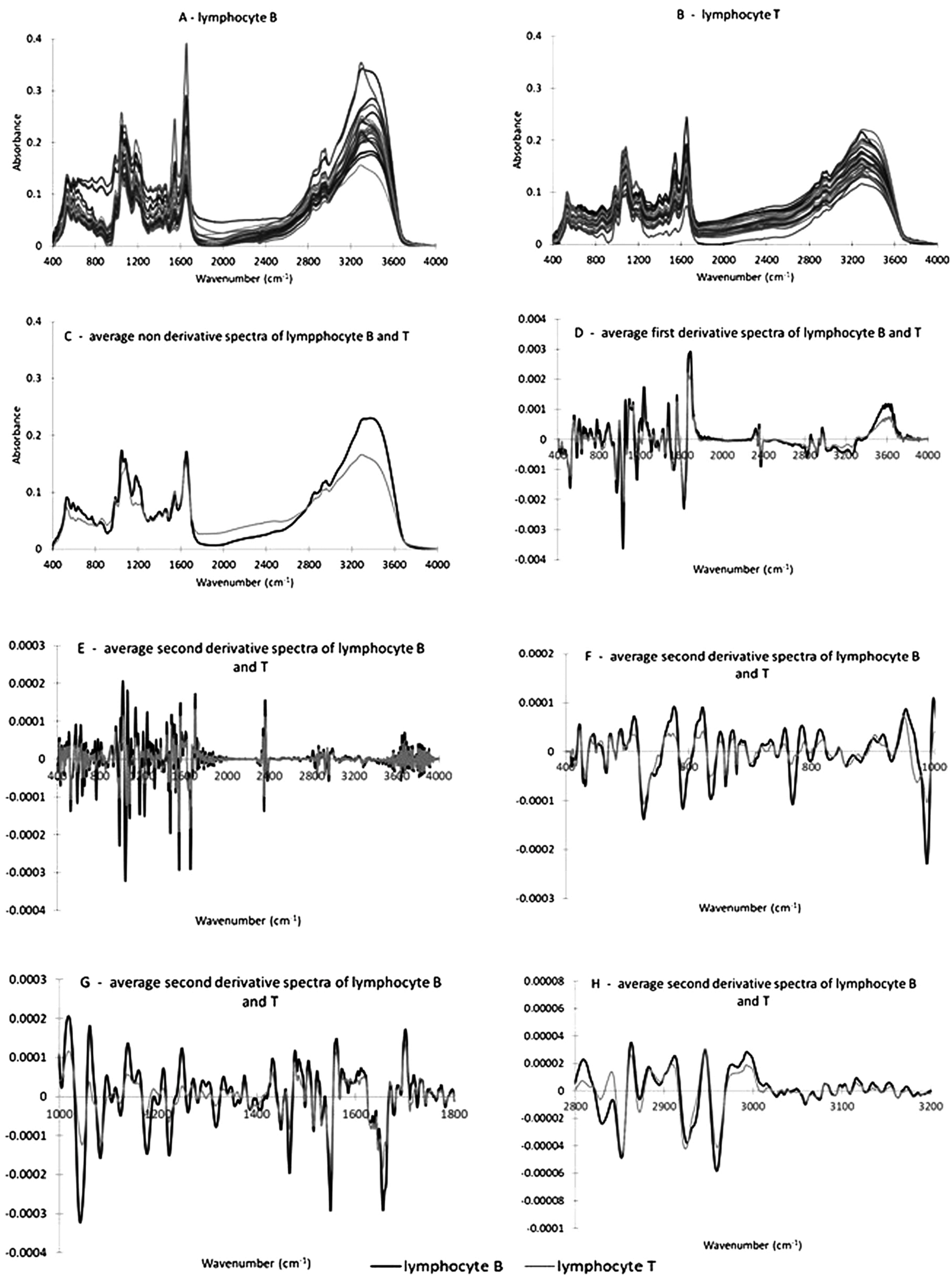

Fig. 1. Spectra of B-lymphocyte (A) and T-lymphocyte (B), and its average non-derivative (C), first (D) and second derivative (E – H) spectra. From C to H graph: Black line, B-lymphocyte; Grey line, T-lymphocyte.

on the use of immune-labeling with expensive monoclonal antibodies, and subsequent laborious processes as based on Immunohistochemistry and flow cytometry [9–12]. For example, the flow cytometry-based process involves several steps, such as the titration of monoclonal antibodies, the equipment compensation of several fluorochromes tagged, and the fixation of samples before running the flow cytometry analysis. Furthermore, the lymphocytes obtained will be marked with antibodies or fixed which could compromise further functional studies.

In the present work, a Fourier-Transformed InfraRed (FTIR) spectroscopic analysis is evaluated to discriminate B and T lymphocytes in a highly specific and sensitive mode, without the use of expensive monoclonal antibodies, i.e. label-free, while enabling a rapid (in the

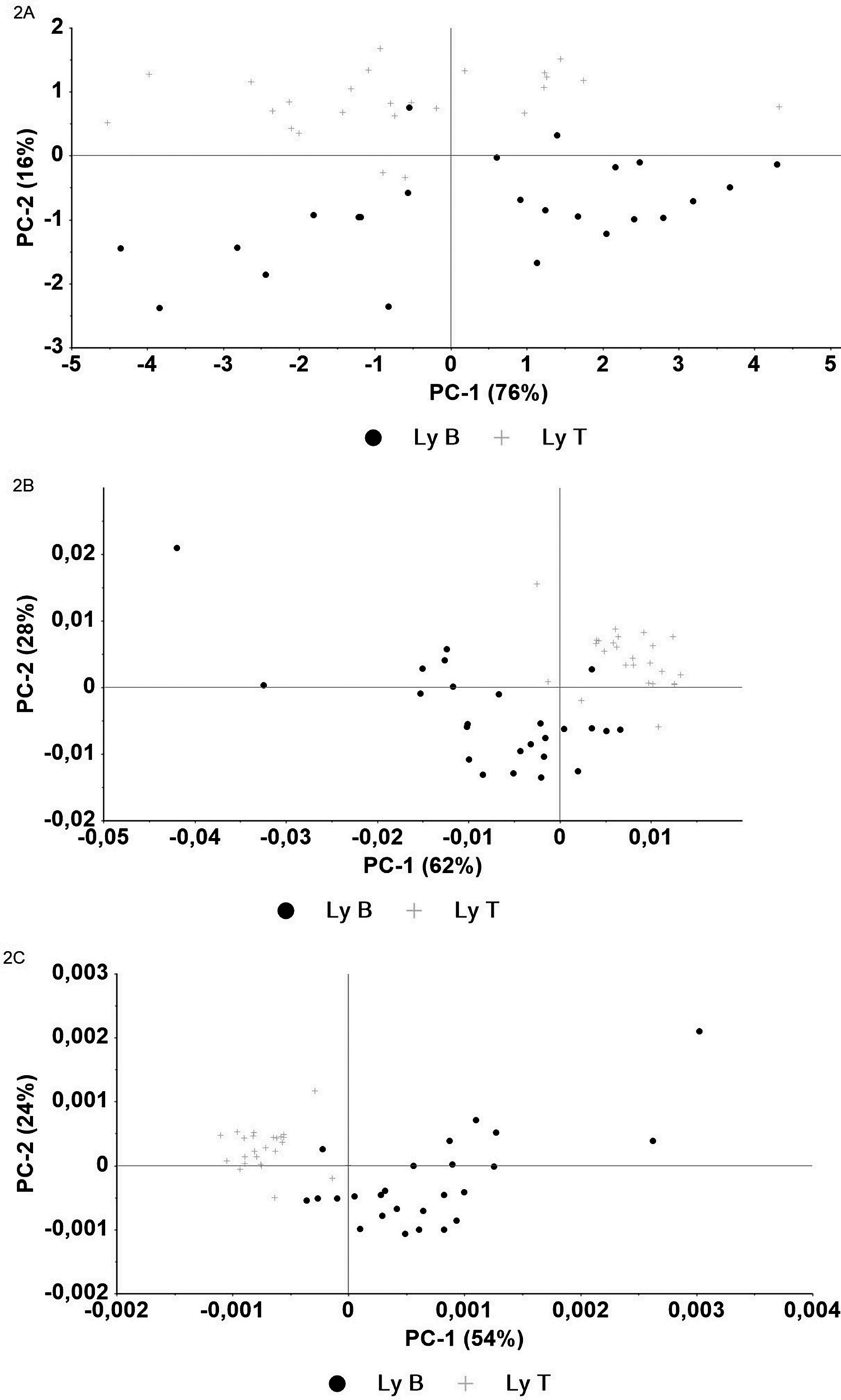

Fig. 2. PCA of lymphocyte spectra with atmospheric correction (A) or with atmospheric correction and first (B) or second derivative (C).,˙ B-lymphocyte; +, T-lymphocyte.

order of minutes), quasi-automatic and high-throughput (e.g. based on microplates) analysis.

FTIR spectroscopy, in the mid infrared region of the spectra, detects fundamental vibrations of chemical groups regarding major molecules present in a cell, in a highly sensitive mode [13–16], being therefore applicable for evaluating biological processes, such as cell infection [16], transfection [15], differentiation [17], mechanisms of cell death [18] and cell discrimination [19–21]. Consequently, FTIR spectroscopy

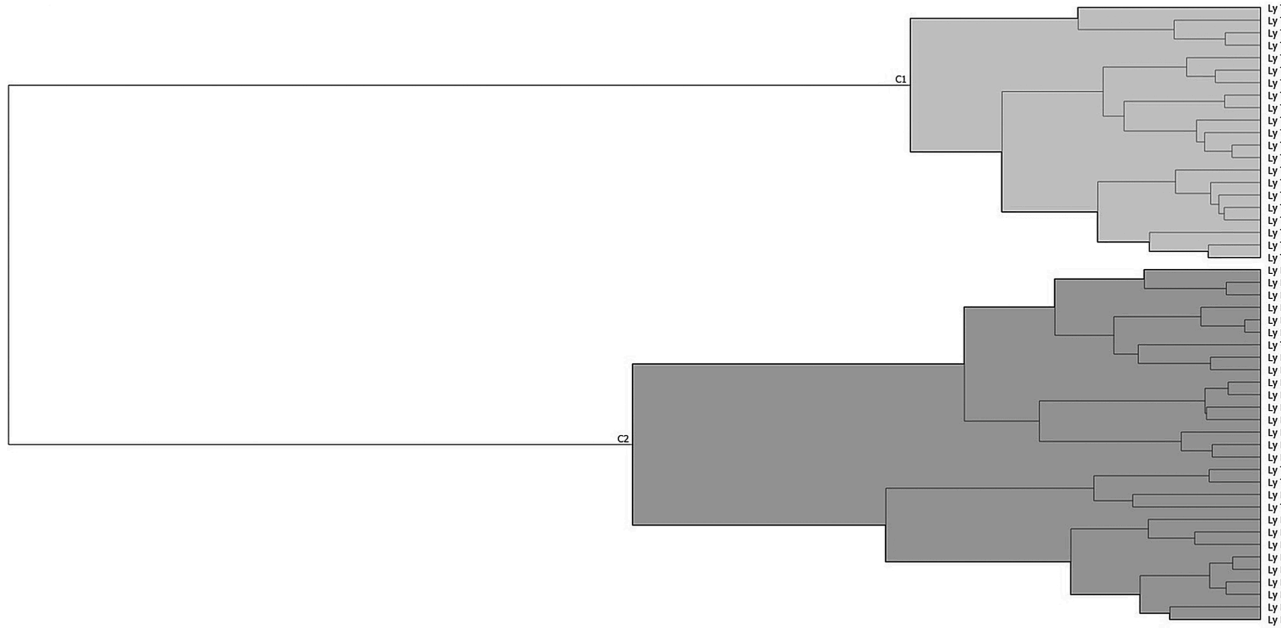

Fig. 3. HCA of lymphocyte spectra with atmospheric correction and first derivative.

has been widely explored either for medical diagnosis, disease prognosis, therapy monitoring, and based on a high diversity of biological samples, such as cells, tissues from biopsies, fluids, e.g. saliva, blood, serum, plasma, and urine [reviewed in [22–27]].

FTIR spectroscopy has already been explored to discriminate B and T-lymphocytes directly on tissue samples of spleen [21] and lymph nodes [20,28]. For example, Wald et al. [20], using FTIR spectroscopy imaging, were able to discriminate by unsupervised (K-means) methods, the spectra of lymph nodes of 14 melanoma patients of the following cells types: B lymphocytes, T lymphocytes, erythrocytes, endothelial and melanoma cells. Subsequently, based on spectra from the 1000−1800 cm^{-1} region, it was possible to discriminate B from T lymphocytes employing a supervised method, namely partial least square discriminant analysis (PLS-DA) from 5 lymph nodes (1 being used for validation and the remain 4 used for testing) [20]. Krafft et al. [21], were also able to discriminate B from T cells in FTIR spectroscopic imaging analysis of tissues obtained from one spleen based on unsupervised analysis (K-means) of spectra between 1800–950 cm^{-1}. Bird et al. [28], were able to discriminate by unsupervised hierarchical cluster analysis (HCA) of micro-FTIR spectroscopic data (based on 900−1800 cm^{-1}) from metastatic lymph nodes of breast cancer, B and T lymphocytes. Despite encouraging results achieved by these researcher groups, and as highlighted by Wald et al. [20], the classification of B and T lymphocytes were conducted in tissues, which are by themselves complex structures. Therefore, the spectral data retrieved from these tissues can include spectral attributes not only from lymphocytes but also from other structures such as small capillaries and other immune cells as macrophages and dendritic cells. Therefore, the classifications achieved could also result from specific mixes of those regions, including e.g. stromal cells. For that reason, Krafft et al. [21] compared spectra retrieved from tissue spleen with one sample from lymphocytes from peripheral blood. Verdonck et al. [29], using isolated B and T lymphocytes from peripheral blood from 6 healthy donors, were able to conduct the correct assignment by PLS-DA of spectra based on 2800–3000 and 1000−1800 cm^{-1} region.

The present work aims to evaluate the discrimination of B and T lymphocytes isolated from peripheral blood from a higher number of healthy donors (n = 25), in relation to the work conducted by Verdonck et al. [29] (n = 6). For that, diverse spectra pre-processing methods were evaluated, followed by its impact on the principal component analysis (PCA) and HCA. It was evaluated the impact of specific spectral regions on B and T-lymphocytes discrimination by HCA. A training and feature-set were used to train a selection algorithm using Support Vector Machine (SVM) classifiers. It was also evaluated, the impact of specific wavenumbers in discriminating these two cell populations.

2. Materials and methods

2.1. Biological assay

10 mL of heparinized sodium whole venous blood were collected from 25 healthy volunteers, which provided their informed consent. Whole blood was diluted at 1:1 (v/v) with RPMI medium (Lonza) and centrifuged at 800 times gravity for 10 min. Buffy coat was extracted and used for B and T-lymphocyte isolation, based on negative sorting with immunomagnetic beads using an EasySep® Kit (StemCell Technologies, Canada), according to the manufacturer's recommendations and resuspended in RPMI medium until usage. Cells were counted in a Neubauer chamber and their viability was verified with trypan blue.

2.2. MIR spectra acquisition

25 μL PBS with 2×10^5 of T or B lymphocytes were plated into a 96-wells Si plate and then dehydrated in a desiccator for 150 min under vacuum. Spectra were collected using an FTIR spectrometer (Vertex 70, Bruker, Germany), equipped with an HTS-XT (Bruker, Germany) accessory. Each spectrum represented 64 coadded scans, acquired in transmission mode between 400 and 4000 cm^{-1}, with a resolution of 2 cm^{-1}. The first well of the 96-wells microplate was left without a sample and the corresponding spectra used as background, according to the HTS-XT manufacturer.

2.3. Spectra pre-processing and processing

All spectra were pre-processed by atmospheric correction (water and carbon dioxide) using OPUS® software, version 6.5 (Bruker, Germany). The first and second derivatives spectra, PCA, HCA, and SVM models were performed in Orange 3 version 3.19.0 (Bioinformatics Lab, University of Ljubljana, Slovenia). First and second derivative spectra were based on a Savitzky-Golay filter, with a 2^{nd} polynomial and a

Table 1
Contingency table of HCA based on lymphocyte spectra with atmospheric correction or with atmospheric correction and first or second derivative over diverse spectral regions.

	400 – 4000 cm^{-1} atmospheric correction			400 – 4000 cm^{-1} atmospheric correction + 1st derivative spectra			400 – 4000 cm^{-1} atmospheric correction+ 2nd derivative spectra		
	C1	C2	∑	C1	C2	∑	C1	C2	∑
Ly B	1	24	25	0	25	25	0	25	25
Ly T	20	5	25	21	4	25	21	4	25
∑	31	29	50	21	29	50	21	29	50
	400 - 1800 cm^{-1} 2nd derivative			900–1800 cm^{-1} 2nd derivative			900 – 1500 cm^{-1} 2nd derivative		
	C1	C2	∑	C1	C2	∑	C1	C2	∑
Ly B	24	1	25	0	25	25	25	0	25
Ly T	1	24	25	20	5	25	1	24	25
∑	25	25	50	20	30	50	26	24	50
	1500 – 1800 cm^{-1} 2nd derivative			400–1000 cm^{-1} 2nd derivative			2700 – 3800 cm^{-1} 2nd derivative		
	C1	C2	∑	C1	C2	∑	C1	C2	∑
Ly B	3	22	25	1	24	25	6	19	25
Ly T	2	23	25	19	6	25	6	19	25
∑	5	45	50	20	30	50	12	38	50

corresponding window size of 15 and 21, respectively. HCA was based on Spearman's rank correlation and hierarchical average-linkage. SVM models used a 3rd degree polynomial kernel with a cross-validation random sampling, based on 80 % and 20 % of data for model training and testing, respectively. The Student's *t* analysis was performed on Microsoft Excel™.

3. Results and discussion

Lymphocytes were isolated from peripheral blood by negative selection through magnetic-activated cell sorting. To obtain spectra with a high signal to noise ratio, B and T lymphocytes were resuspended at ~2 × 10^5 cells per 25 μL in phosphate buffer. Despite the use of a similar number of cells in all the assays, highly variable absorbances were observed among the different spectra (Fig. 1A, 1B), indicating that each sample presented a slightly different number of cells. For example, the absorbance at ~3300 cm^{-1} of the B lymphocyte samples varied between 0.16 and 0.35, whereas absorbances from T lymphocyte varied between 0.12 and 0.22. The average spectrum of B and T lymphocyte is represented in Fig. **1C**, with already apparent differences found on spectra regions between 400 cm^{-1} and 1300 cm^{-1} and between 2800 and 3600 cm^{-1}. To resolve superimposed bands, while minimizing artifacts such as physical phenomena, i.e. emphasizing biochemical information, the average of first (Fig. 1D) and second derivative spectra were also analyzed (Fig. 1E to 1 H). Derivatives increase band resolution, which is particularly relevant in complex biological samples, such as cells, resulting in diverse overlapping bands [13].

It was observed in the PCA score plot, a separation between spectra of B and T lymphocyte, even based on spectra pre-processed only for atmospheric compensation (Fig. 2). The PCA based on derivative spectra improved this separation. As expected, derivatives led to the reduction of variance represented in the first PCs, which however still presented a high data variance (Fig. 2). Although, apparently, there is not a clear major difference among these cells under the optical microscope [30], they do differ among different spectra features, according to their molecular specificities.

The dendrogram of HCA, based on spectra with atmospheric correction and the first or second derivative, always presented two clusters, with good separation between B and T-lymphocytes (Fig. 3, Table 1). HCA based on non-derived spectra resulted in only 6 misclassified samples (out of 50, representing 12 % of samples), where one B lymphocyte sample was misclassified in the T lymphocytes group, and 5 T cells were misclassified in the B cells group. Both first and second derivative spectra reduced the number of misclassified samples from 6 to 4, whereas all B cells were correctly classified, and 4 T lymphocyte were wrongly classified on the B cell cluster. Interestingly, derivative spectra resulted in a gain on cell classification, i.e. in discriminating B from T lymphocyte, albeit at the expense of a decrease in distance between the two clusters of cells (data not shown), probably due to noise

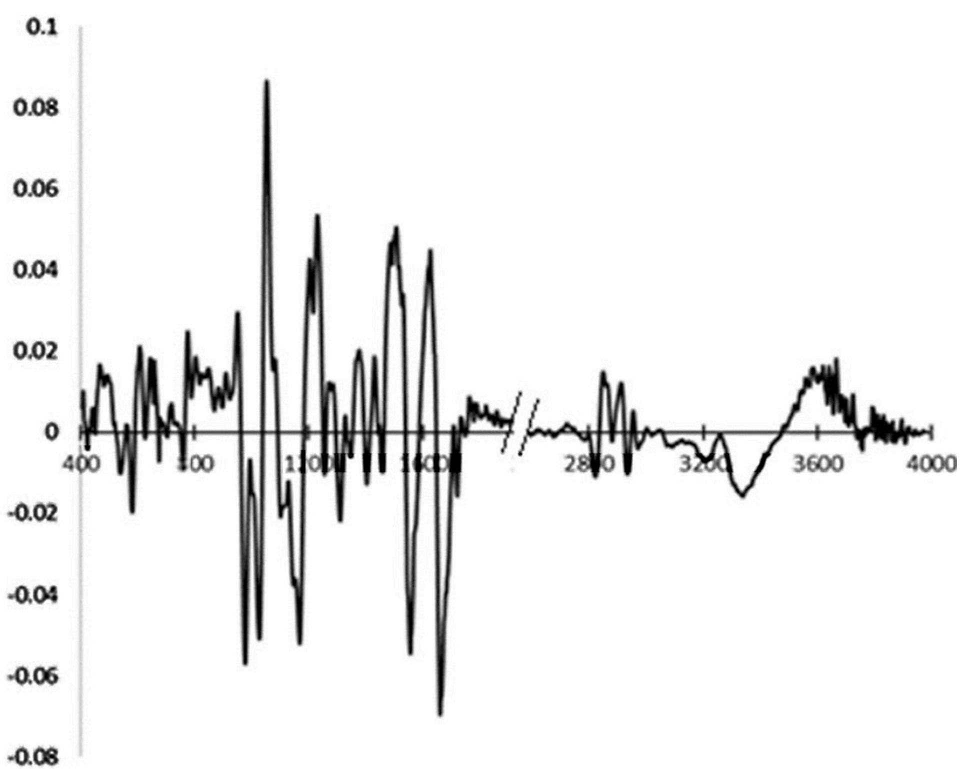

Fig. 4. PC1 loading of PCA based on second derivative spectra as represented in Fig. 2C.

Table 2
Predictive characteristics of SVM models-based on lymphocyte spectra with atmospheric correction or with atmospheric correction and first or second derivative over diverse spectral regions.

Spectra region	AUC	Accuracy	F1	Precision	Recall
atmospheric correction					
400−4000 cm^{-1}	0.96	0.82	0.82	0.85	0.82
atmospheric correction + 1st derivative spectra					
400−4000 cm^{-1}	0.99	0.90	0.90	0.92	0.90
atmospheric correction + 2nd derivative spectra					
400−4000 cm^{-1}	1.00	0.95	0.95	0.95	0.95
400−1800 cm^{-1}	1.00	0.95	0.95	0.95	0.95
900−1500 cm^{-1}	1.00	0.95	0.95	0.96	0.95

amplification. The first derivative and second derivative resulted in a decrease in the distance between the two clusters of 19 % and 48 % comparatively to HCA based on the non-derived spectra.

The HCA based on the second derivative spectra resulted in the same number of misclassified samples as based on the first derivative, most likely as a result of the balance between the gain in captured information, due to the increase in the order of the derivative, and noise amplification. An advantage of the second derivative is that the spectra are not affected by the number of cells present in the assay, and consequently, the classification obtained is most probably due to the biochemical characteristics of the cells and not small differences in cell quantity used for spectra acquisition.

To evaluate the spectral region that best contributed to B and T lymphocyte discrimination, it was considered the PC1 loadings based on second derivative spectra (Fig. 4). The most significant spectral regions that contributed to discrimination between B and T-lymphocytes, were between 400 to 1800 cm^{-1} and between 2700 and 3800 cm^{-1}. The effect of the regions and sub-regions of the second derivative spectra were evaluated by HCA. The 400 to 1800 cm^{-1} spectral region resulted in improved discrimination of HCA between B and T-lymphocytes in relation to whole spectra, which led to the reduction of sample misclassification from 4 to 2 (Table 1). This is according to other authors that were able to discriminate B from T lymphocytes based on a region between 900 to 1800 cm^{-1}, as observed by Wald et al. [20], Krafft et al. [21], and Bird et al. [28]. In the present work, the second derivative between 900 to 1800 cm^{-1}, resulted in 5 misclassified samples. The HCA that resulted in only 1 misclassified T-lymphocytes sample, was based on a region between 900 to 1500 cm^{-1} (Table 1).

Diverse predictive SVM models of B and T lymphocytes based on spectra were also developed (Table 2), which make use of spectra close to the class borders, as support vectors, to define a discriminant surface [31]. The best models were based on derivative spectra, in which SVM models presented a high accuracy (0.95), precision (0.95), recall (0.95), and F1 score, i.e. the weighted average of precision and recall (0.95). Concerning the area under the curve (AUC), this model presented a value close to 1, i.e. the model enabled to maximize true positive rates (i.e. the recall) while minimizing the false-positive rates. SVM models based on sub-regions of the second derivative spectra, which also resulted in HCA with high classifications (i.e. between 400 to 1800 cm^{-1} and between 900 to 1500 cm^{-1}), maintained the high predictivity obtained in SVM models based on whole spectra (i.e. between 400 to 4000 cm^{-1}).

To evaluate the most relevant spectral peaks that discriminate B from T lymphocytes, it was considered the negative peaks of second derivative spectra, as it corresponds to normal peaks at the non-derived spectra, while resulting in sharper peaks, easier to identify and consequently to analyze. To avoid the identification of peaks due to noise amplification, peaks that were present in most of the second derivative spectra and that correspond to peaks or shoulders at the non-derived spectra were selected. A Student's *t*-test was applied to compare the second derivative absorbances at these wavenumbers (i.e. peak height) between all spectra from B cells with spectra from T lymphocytes. From

Table 3
Average values and standard deviations of B and T-cells of absorbances of negative peaks of second derivative spectra, and *p*-value of student's *t*-test regarding the comparison of B and T-cells population.

Wavenumber (cm^{-1})	B cells		T-cells		*p*-value
	Average	Standard deviation	Average	Standard deviation	
526	−1.38E-04	2.01E-08	−1.07E-04	1.31E-08	1.14E-04
591	6.27E-05	7.49E-09	2.81E-05	7.70E-09	1.21E-08
635	−9.73E-05	4.48E-09	−5.09E-05	4.06E-09	1.86E-16
768	−1.08E-04	1.81E-08	−2.19E-05	5.91E-09	3.23E-16
855	3.42E+01	7.02E+05	−3.16E-05	1.50E-09	3.27E-01
931	−1.86E-05	1.46E-09	−9.16E-06	4.60E-10	5.38E-06
987	−2.29E-04	8.34E-08	−1.04E-04	2.40E-08	2.74E-11
1042	−3.23E-04	1.33E-07	−1.02E-04	5.22E-08	1.55E-15
1082	−1.57E-04	2.83E-08	−1.09E-04	9.96E-09	5.08E-07
1126	−4.07E-05	6.25E-09	−8.38E-06	2.60E-09	1.80E-10
1176	−1.43E-04	2.78E-08	−4.59E-05	9.27E-09	6.79E-15
1222	−1.51E-04	2.31E-08	−6.49E-05	7.50E-09	1.28E-14
1262	−3.81E-05	8.76E-09	6.10E-06	3.48E-09	3.14E-12
1281	−1.45E-05	8.00E-10	−5.81E-06	1.55E-10	7.96E-08
1316	−7.43E-05	9.52E-09	−2.36E-05	3.95E-09	1.97E-13
1446	6.93E-05	4.07E-08	4.47E-05	2.04E-08	1.89E-02
1466	−1.97E-04	1.81E-07	−8.16E-05	8.12E-08	2.05E-06
1549	−2.93E-04	1.04E-06	−1.92E-04	6.57E-07	6.43E-02
1655	−2.92E-04	1.20E-06	−1.83E-04	7.95E-08	2.59E-02
1725	−2.76E-05	2.93E-07	5.32E-06	1.44E-07	2.29E-01
2852	−4.84E-05	4.37E-09	−4.52E-05	3.42E-09	3.80E-01
2871	−6.15E-06	9.60E-09	−1.54E-05	1.99E-09	4.28E-02
2926	−3.74E-05	9.88E-09	−3.75E-05	2.95E-09	9.86E-01
2959	−5.82E-05	2.18E-08	−4.13E-05	3.54E-09	1.39E-02

the 24 peaks evaluated, 15 presented values statistically different ($p < 0.001$) (Table 3). Ten out of the 15 significant peaks were included in the region between 900 and 1500 cm^{-1}, which resulted in the best B and T lymphocytes discrimination by HCA. Regarding the significant peaks ($p < 0.001$), some were found to be due to CH_3 vibrations from lipids such as at 1466 and 1316 cm^{-1}, phosphate groups from lipids, phosphoproteins, and nucleic acids at 1082 and 1262 cm^{-1}, from nucleic acids at 1126 and 1176 cm^{-1}, whereas the remaining were from the fingerprint region, representing bond vibrations of highly diverse biomolecules. Box plots of second derivative absorbances, at some of the peaks that resulted in the lowest *p*-values, are represented in Fig. 5.

By analyzing the PCA score plot based on derivative spectra (Fig. 2), it was observed that B cells samples were more dispersed in the score plot than samples from the T-lymphocyte, pointing to an apparent higher diversity of molecular fingerprint among all B cell samples when compared to T-lymphocytes. This apparent higher diversity of molecular profiles among B cells was also corroborated in the HCA dendrogram

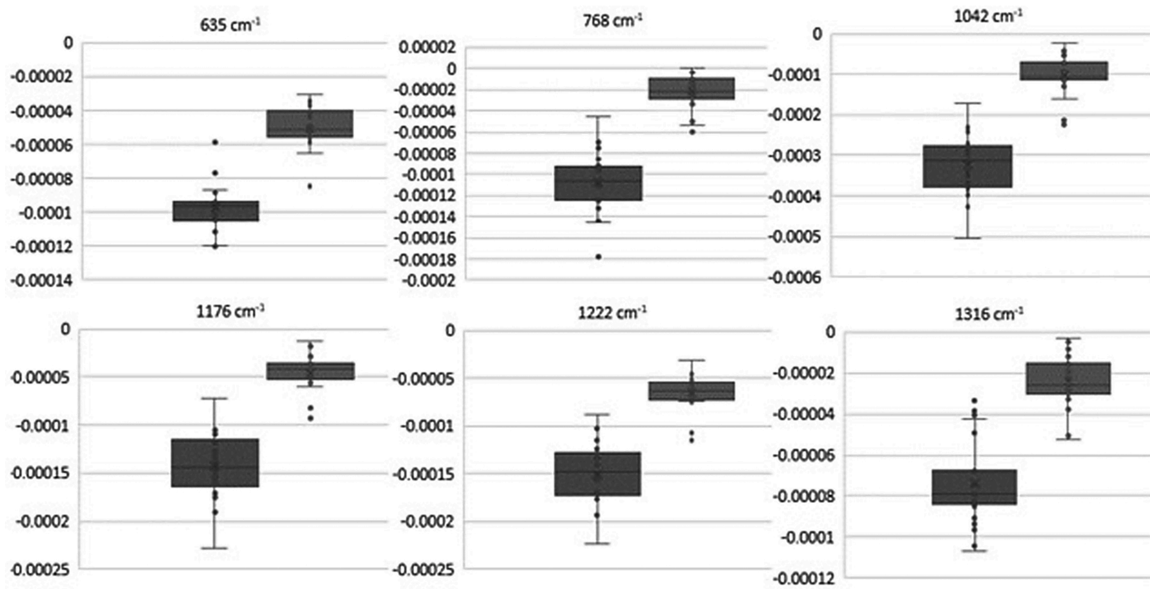

Fig. 5. Box plot of absorbances of negative peaks of second derivative spectra of B-lymphocyte (left panels) and T-lymphocyte (right panels).

based on second derivative spectra (Fig. 3), as T lymphocytes are found at closer distances between them in relation to distances between B cells. The B cell population also presented higher standard deviations of second derivative absorbances at the 24 wavenumbers analyzed in relation to the T-lymphocyte population (Table 3).

4. Conclusions

It is important to be able to identify the dominant lymphocyte population (B or T) in diverse pathological situations. However, conventional methods to quantify these lymphocyte populations are usually expensive and time-consuming. In the present work, it was evaluated the discrimination on these two cell populations by FTIR spectroscopy, considering 25 samples of T and 25 samples of B cells from healthy donors. The FTIR spectroscopic analysis was based on a simple (i.e. based on dehydrated cells), rapid (a single spectrum is acquired around the one-minute mark), label-free, economic (as no reagents are needed), and on a high-throughput mode (i.e. based on microplates with multiwells). It was observed that the second derivative spectra led to an increase in cell discrimination by HCA, by which only 4 out of 50 samples were misclassified, in relation to non-derived spectra. Based on PC1 loadings on the second derivative spectra, it was possible to identify spectral regions that most contributed to the B and T lymphocytes separation in the PCA score plot. The impact of these sub-regions was evaluated by HCA. The second derivative based on 900 to 1500 cm^{-1} resulted in one unique T-lymphocyte sample being misclassified out of 50 samples. This last spectral region (i.e. between 900 to 1500 cm^{-1}) presented diverse peaks, that, in mean terms, were significantly different ($p < 0.001$) between B and T cells, and that were associated with lipids, phosphate groups, and nucleic acids. The PCA, HCA, and analysis of absorbance at specific wavenumbers revealed that that B cells presented a higher diversity of molecular profile in relation to T lymphocytes. A SVM model enabled to predict B and T lymphocyte from the molecular profile acquired by the second derivative spectra with high accuracy (0.95), precision (0.95), recall (0.95), and F1 score (0.95). FTIR spectroscopy presents therefore, appealing characteristics for a new diagnostic tool to analyze and discriminate B from T-lymphocytes, paving the way for newer approaches in the study of B and T-lymphocytes.

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Research ethics

We further confirm that any aspect of the work covered in this manuscript that has involved human patients has been conducted with the ethical approval of all relevant bodies and that such approvals are acknowledged within the manuscript. Written consent to publish potentially identifying information, such as details or the case and photographs, was obtained from the patient(s) or their legal guardian(s).

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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